

ALLIANCE UNIVERSITY

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ALLIANCE UNIVERSITY

Alliance School of Advanced Computing

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Project Title:

Project Phoenix – AI-Powered Student Project & Career Management System

Signature of Student

Signature of Faculty & Project Guide

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1. Abstract

Project Phoenix – AI-Powered Student Project & Career Management System is a unified academic platform designed to connect project assignment, student submission, evaluation, analytics, skill development, and career guidance in one workflow. The project addresses three major challenges identified in the project concept: project work being scattered across platforms, feedback being delayed or inconsistent, and students being uncertain about suitable career pathways.

The system provides role-based workspaces for students, teachers, and parents. Teachers can create project assignments, define requirements and rubrics, review submissions, provide final marks and feedback, and view class-level insights. Students can view assigned work, submit projects, monitor progress, build portfolios, earn points and badges, and explore career pathways. Parents can view linked student progress and participate in PTM-oriented career discussions.

An AI-assisted evaluation layer provides a preliminary assessment of project quality, completeness, accuracy, and areas for improvement. The final grade remains under teacher review. The dataset used in this project contains 1,000 student-project records and supports analysis of project performance, skills, submission behaviour, AI feedback, career interests, and recommended careers.

The resulting platform is intended as an academic prototype that combines project management with data-driven student development rather than treating grades as the only measure of progress.

2. Introduction

Project-based learning gives students an opportunity to apply classroom concepts to practical problems. However, when project assignment, submission, evaluation, feedback, and career guidance are handled separately, students may lose visibility of their progress and teachers may spend significant time organizing information.

The Project Phoenix concept proposes a connected learning journey. The project presentation describes a centralized hub for project assignment and management, AI-assisted evaluation, career exploration, gamification, portfolio development, and parent-teacher counselling. The workflow is designed around a human-in-the-loop model: AI performs an initial assessment while teachers provide contextual judgment and publish the final result.

The interface design organizes the system into separate student, teacher, and parent workspaces. Student navigation includes Dashboard, My Projects, Career Guidance, Portfolio, Leaderboard, and Messages. Teacher navigation includes Dashboard, Assignments, Evaluation Inbox, Class Insights, PTM Counselling, and Messages. Parent navigation includes Overview, Student Progress, PTM Counselling, Goals & Pathways, and Messages.

3. Problem Statement

The project is motivated by the following practical problems:

- Lack of centralization: project assignments and progress information may be distributed across different tools, making tracking difficult.
- Limited feedback: students may receive delayed or inconsistent feedback, reducing the usefulness of project work for learning.
- Career uncertainty: students may not clearly understand how their project skills relate to possible career paths.
- Disconnected evaluation: project submission, AI feedback, teacher grading, and student reflection can become separate activities.
- Limited progress visibility: students, teachers, and parents need different views of the same learning journey.
- Weak evidence of skills: grades alone do not show the projects, competencies, feedback, and achievements a student has built.

The project presentation explicitly frames the original problem around lack of centralization, limited feedback, and career indecision.

4. Existing System

In a conventional academic workflow, project instructions may be shared through separate classroom or communication platforms, students submit files through another channel, teachers maintain marks separately, and career counselling may occur independently. Such a fragmented workflow makes it harder to connect a student's project activity with skill development and future goals.

Major limitations of the existing approach include:

- No single dashboard showing assignments, deadlines, submissions, feedback, and progress.
- Manual effort for teachers to organize submissions and compare student performance.
- Feedback may not be standardized across projects.
- Career guidance is often separated from evidence of actual project performance.
- Student portfolios and achievements may not be connected to teacher-reviewed work.
- Parents may receive only periodic grade information instead of a broader progress picture.

The Project Phoenix concept responds to these gaps through centralized project management, AI-assisted evaluation, career exploration, points, leaderboards, portfolios, and PTM counselling.

5. Proposed System

Project Phoenix proposes a role-based web platform that connects the complete project lifecycle. The core workflow is: teacher assignment → student submission → AI preliminary

evaluation → teacher review → final grade and feedback → analytics, portfolio evidence, and career guidance.

5.1 Main Modules

Student Module: Dashboard, assigned projects, submission, progress tracking, portfolio, points/badges, leaderboard, career guidance, and messages.

Teacher Module: Assignment creation, transparent rubric definition, submission/evaluation inbox, project checking, marks and feedback, class insights, PTM counselling, and messages.

Parent Module: Student progress, reviewed results, goals and pathways, PTM counselling, and notifications.

Project Management: Project title, subject, class/group, skill level, due date, requirements, accepted file types, project links, and rubric-based assessment.

AI-Assisted Evaluation: Preliminary checks for completeness, accuracy, rubric alignment, strengths, and improvement suggestions. The AI result is not the final grade.

Career Guidance: Career exploration based on skills and interests, with suggested pathways and next skills to practice. Recommendations are guidance rather than guaranteed outcomes.

Gamification & Portfolio: Teacher-reviewed points, class leaderboard, badges, and a portfolio containing selected reviewed projects and demonstrated skills.

5.2 Project Workflow

1. Teacher creates an assignment with instructions, requirements, skill level, deadline, and assessment rubric.
2. Student views the assignment, works on the project, uploads the required files or project link, and submits the work.
3. The system performs a preliminary AI-oriented evaluation of completeness, accuracy, and rubric alignment.
4. The teacher opens the evaluation inbox, checks the original student work and AI observations, assigns final rubric scores, and adds feedback.
5. The published result becomes the student's final reviewed grade and contributes to points, portfolio evidence, analytics, and progress tracking.
6. Career guidance uses project and skill evidence together with stated interests to present exploratory recommendations.

The project presentation describes the same sequence as Teacher Assignment, Student Submission, AI Evaluation, and teacher-focused feedback. The detailed UI reference also states that AI scores are preliminary and teacher review determines the final grade.

6. About Dataset

The cleaned Project Phoenix dataset is a structured student-project assessment dataset. It contains 1,000 records and 21 columns. Each record links a student with demographic and education information, a project, competency scores, submission status, project score, points

earned, AI feedback, career interest, and recommended career. The dataset therefore supports both academic-performance analysis and career-guidance analysis.

Dataset Property	Value
Records	1,000 student-project records
Columns	21
Unique Project IDs	100
Project Subjects	10
Cities	10
Skill Levels	Beginner, Intermediate, Advanced
Education Levels	School, Higher Secondary, Undergraduate, Postgraduate
Submission Status	Submitted on time / not submitted on time
Assessment Areas	Programming, analytical, creativity, communication, problem solving, completeness, accuracy
Guidance Fields	AI feedback, career interest, recommended career

The dataset documentation describes the same structure: student identity and education attributes, project subject and skill level, seven scored competencies, submission status, project score, points earned, AI feedback, career interest, and recommended career.

6.1 Important Dataset Fields

Field Group	Fields	Purpose
Student	student_id, student_name, age, city, education_level	Identifies and profiles the learner.
Project	project_id, project_subject, skill_level	Describes the assigned project and expected level.
Competencies	programming_score, analytical_score, creativity_score, communication_score, problem_solving_score, completeness_score, accuracy_score	Measures project-related competencies.
Performance	submitted_on_time, project_score, points_earned	Tracks completion and achievement.
AI/Feedback	ai_feedback	Stores preliminary qualitative feedback.
Career	career_interest, recommended_career	Compares learner interest with recommended pathway.

6.2 Dataset Analysis

Analysis of the cleaned dataset shows an average project score of 69.79 and an average of 94.33 points earned. 808 records were marked as submitted on time, while 192 were not.

Intermediate-level projects form the largest skill group with 458 records, followed by Beginner with 401 and Advanced with 141.

The most frequent project subjects are Business Analytics (110), Database (107), Cyber Security (105), Data Science (103), and UI/UX Design (103). The average competency scores are highest for completeness (77.30), followed by problem solving (67.08) and communication (65.01).

In the career recommendation field, UI/UX Designer is the most frequent recommendation with 277 records, followed by Product Manager with 213, Mobile App Developer with 117, Database Administrator with 107, and Data Analyst with 93.

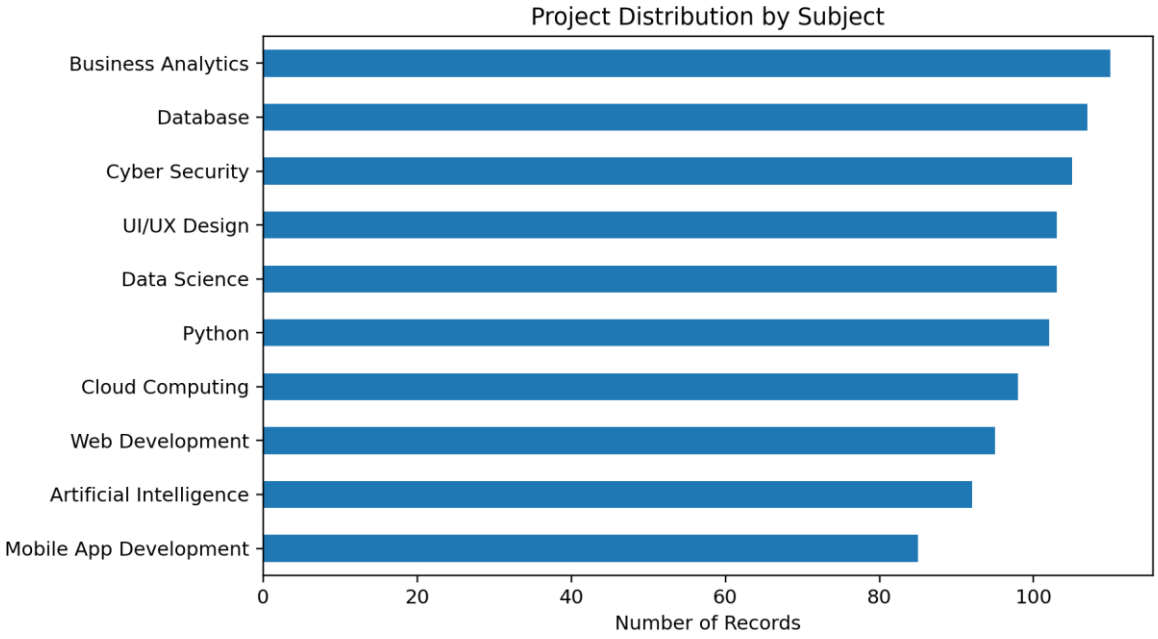


Figure 1. Distribution of student projects by subject

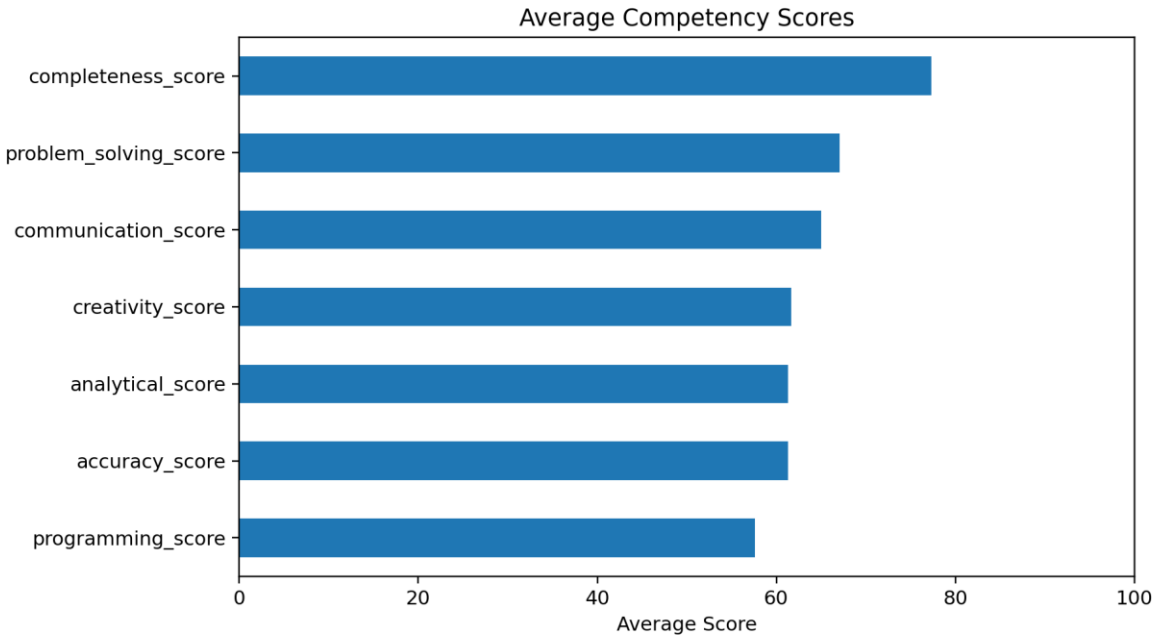


Figure 2. Average competency scores in the dataset

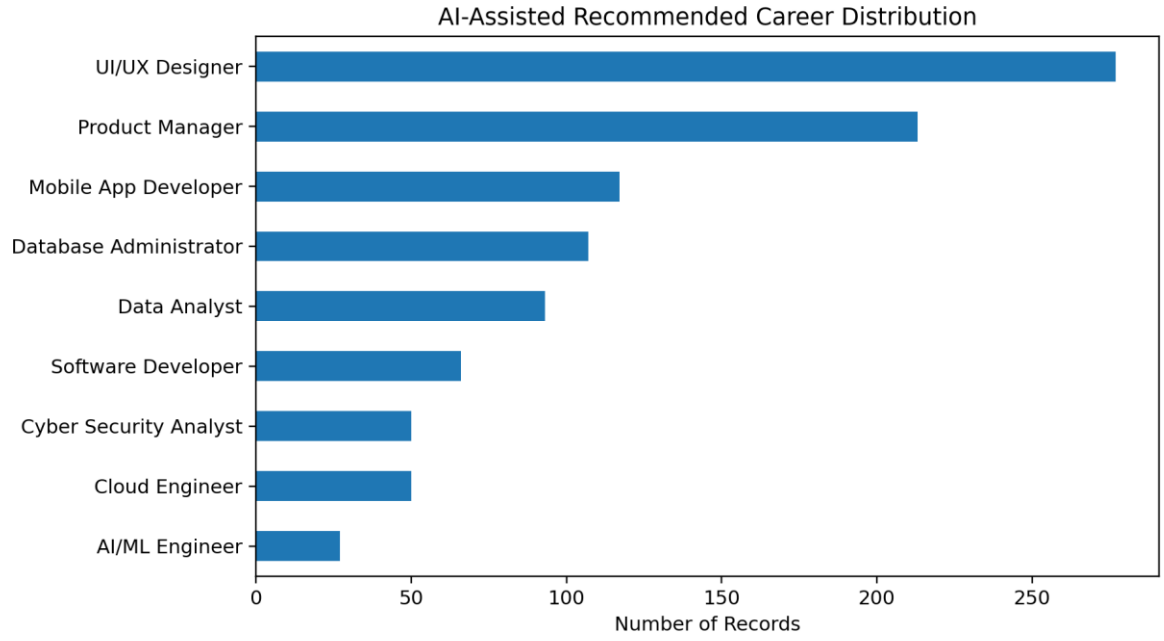


Figure 3. Distribution of AI-assisted recommended careers

7. Technologies Used

The project combines web technologies, application logic, structured data, and analytics. The following stack is suitable for the implemented Project Phoenix prototype and the dataset-driven dashboard workflow:

Technology / Tool	**Role in Project**
HTML5	Structures the student, teacher, parent, assignment, evaluation, dashboard, and career-guidance pages.
CSS3	Provides responsive layout, cards, tables, forms, navigation, status styles, and visual hierarchy.
JavaScript	Handles client-side interaction, form behaviour, dynamic UI updates, validation, and dashboard interactions.
Python / Flask	Provides application-side routing, business logic, data handling, and integration between the interface and stored project data.
Pandas	Supports dataset loading, cleaning, filtering, aggregation, and analytical preparation.
CSV Dataset	Stores the Project Phoenix student-project assessment data used by the analytics and recommendation views.
Power BI	Supports visual analysis of project performance, skill scores, submission behaviour, and career-related distributions.

Technology / Tool	Role in Project
Web Browser	Provides the execution environment for the responsive web interface.

The UI reference specifies a responsive desktop-first interface with a 12-column grid and role-specific navigation, together with reusable cards, tables, forms, project status indicators, and dashboard patterns.

8. Data Flow Diagram

The Level-1 DFD below represents the major information flow in Project Phoenix. Students and teachers interact with the application for project work and evaluation, parents access progress and counselling information, the application stores project/user data, and the AI/analytics layer produces evaluation and career/performance insights.

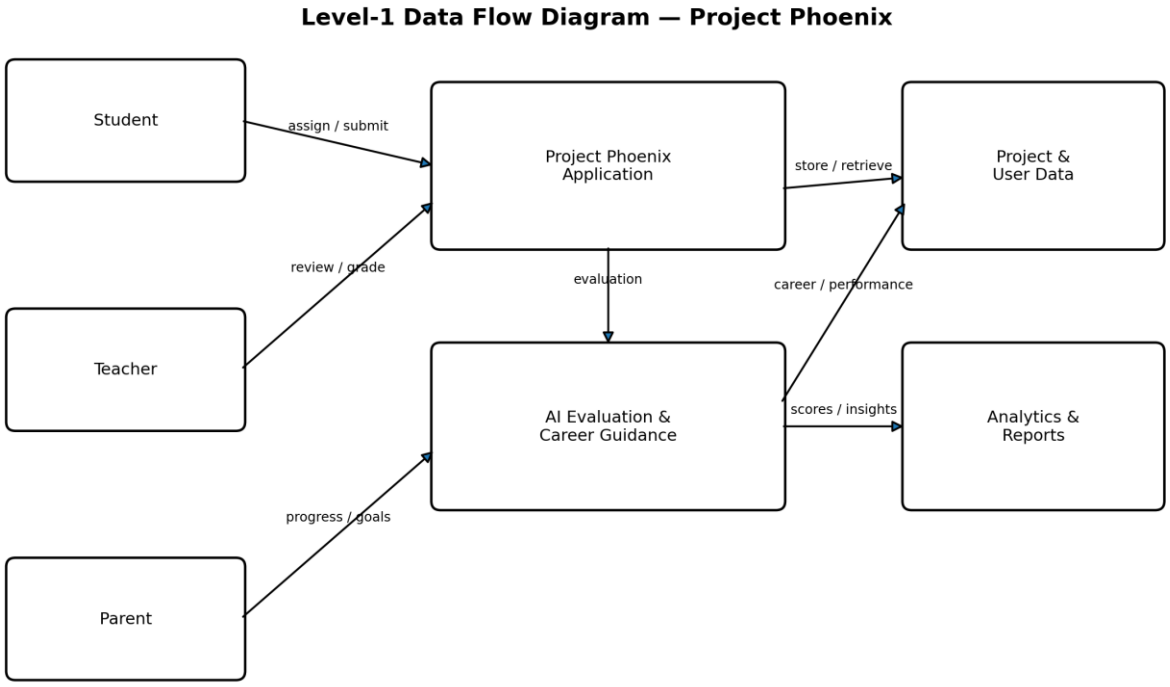


Figure 4. Level-1 Data Flow Diagram of Project Phoenix

8.1 DFD Explanation

- Student → Application: views assignments, submits project work, reviews feedback, and tracks progress.
- Teacher → Application: creates assignments, reviews submissions, enters final marks, and publishes feedback.
- Parent → Application: views linked progress and participates in PTM/career planning.

- Application ↔ Project/User Data: stores and retrieves student, project, assessment, and progress information.
- Application → AI Evaluation & Career Guidance: sends relevant project/skill information for preliminary evaluation and pathway suggestions.
- Analytics & Reports: converts stored assessment data into class insights, charts, performance summaries, and career distributions.

8.2 Detailed System Architecture

Project Phoenix can be understood as a layered academic application. The project presentation and interface reference describe role-specific workspaces connected through a shared project workflow. The layers below are a logical description of the proposed prototype, not a claim that every production service has already been deployed.

Layer	Responsibilities	Typical Information
Presentation	Displays role-specific dashboards, forms, tables, project cards and status indicators.	Assignments, deadlines, progress, review status and notifications.
Application	Coordinates assignment creation, submission handling, evaluation workflow and role-based views.	Project rules, submission state, rubric criteria and feedback.
Data	Stores and retrieves student, project, assessment and progress records.	Student/project records, competency scores, points and career fields.
Analytics	Aggregates records for class insights, charts and performance summaries.	Subject distribution, competency averages, submission rates and career counts.
AI assistance	Provides preliminary completeness, accuracy and rubric-alignment observations.	Draft evaluation notes and improvement suggestions for teacher review.

A layered structure makes responsibilities easier to understand and maintain. The presentation layer focuses on usability, application rules control how records move between assignment, submission, review and publication, analytics summarizes stored records, and AI assistance supports rather than replaces teacher judgement.

The student workspace focuses on completing assigned work and understanding progress.

The teacher workspace focuses on assignment definition, review, grading and class-level insight.

The parent workspace focuses on progress visibility, goals and counselling conversations.

Shared records allow one learning activity to support assessment, portfolio evidence and career exploration.

8.3 Functional Requirements

Functional requirements describe the actions the platform should support. They are organized by user role and shared workflow so that expected behaviour can be checked during development and demonstration.

ID	Requirement	Expected Behaviour
FR-01	Role-specific access	Students, teachers and parents see the workspace relevant to their role.
FR-02	Assignment creation	Teachers define title, subject, level, instructions, deadline, accepted files and rubric.
FR-03	Student submission	Students upload permitted files or provide project links and submit work.
FR-04	Submission status	Draft, submitted, under review and reviewed states are understandable.
FR-05	AI preliminary review	AI observations are labelled preliminary and distinguished from final marks.
FR-06	Teacher grading	Teachers inspect original work, assign rubric scores and add feedback.
FR-07	Grade publication	Only a teacher-published result is treated as the final reviewed grade.
FR-08	Student progress	Students see reviewed results, deadlines, points, badges and project evidence.
FR-09	Career exploration	Students compare interests and demonstrated skills with exploratory pathways.

FR-10	Parent progress view	Parents see permitted progress information for their linked student.
FR-11	Class analytics	Teachers view aggregated class indicators and competency distributions.
FR-12	Portfolio	Students collect teacher-reviewed projects and related skill evidence.

A prototype demonstration should show the complete path: a teacher creates an assignment, a student submits it, a preliminary evaluation appears, the teacher reviews and publishes the result, and the student dashboard reflects the reviewed outcome.

8.4 Non-Functional Requirements

Non-functional requirements describe the qualities expected from the platform. They matter because the application contains academic records, assessment comments and information used in career conversations.

Quality	Requirement and Rationale
Usability	Navigation labels and status messages should be understandable to users with different levels of technical confidence.
Consistency	Similar actions such as viewing a project, checking a deadline and reading feedback should use consistent layouts.
Performance	Dashboards should avoid unnecessary repeated calculations and summarize large tables where practical.
Reliability	Submitted work and published grades should remain associated with the correct student and assignment.
Privacy	Each role should receive only information needed for its responsibilities.
Transparency	AI observations, teacher feedback and final marks should be visibly differentiated.
Maintainability	Modules should have clear responsibilities so project management, evaluation and analytics can be improved separately.
Accessibility	Forms should have clear labels, readable contrast and meaningful status text.

Portability	Accepted submission formats and export options should be stated clearly.
Auditability	Important changes to submissions, marks and publication state should be traceable in a future production version.

These requirements are design goals and evaluation criteria. Their inclusion does not mean that every item has been independently verified in a deployed production environment.

8.5 Use-Case Descriptions

Use Case 1: Teacher Creates an Assignment

Primary actor: Teacher. Preconditions: the teacher has access to the teacher workspace. Main flow: the teacher opens Assignments, enters the project title and subject, specifies instructions and a due date, chooses accepted submission formats, defines rubric criteria, checks the details and publishes the assignment. Expected result: the assignment is available in the relevant student project list.

Use Case 2: Student Submits Project Work

Primary actor: Student. Preconditions: an assignment is available. Main flow: the student opens My Projects, reads requirements, prepares the report or artefacts, uploads permitted files or adds a project link, checks the submission and submits it. Expected result: a submission is associated with the correct assignment and a visible status.

Use Case 3: Teacher Reviews and Publishes a Grade

Primary actor: Teacher. Preconditions: a student submission exists. Main flow: the teacher opens the Evaluation Inbox, reviews the submission and preliminary AI observations, scores the work against the rubric, writes feedback and publishes the final grade. Expected result: the reviewed result becomes available in the student progress view.

Use Case 4: Student Explores Career Pathways

Primary actor: Student. Preconditions: project and skill information is available. Main flow: the student reviews demonstrated skills and stated interests, examines suggested pathways, reads the explanation and identifies skills to practise next. Expected result: exploratory next steps rather than a guaranteed career prediction.

Use Case 5: Parent and Teacher Discuss Progress

Primary actors: Parent and teacher. Preconditions: the parent is linked to the student and relevant progress records are available. Main flow: both parties review project evidence, strengths, improvement areas and goals. Expected result: a more informed counselling conversation based on reviewed work.

These use cases describe the intended workflow and provide scenarios that can be used in a demonstration or future acceptance test.

8.6 Assignment Rubric and Submission Rules

The project concept includes a rubric that makes evaluation criteria visible before work begins. Transparent criteria help students plan submissions and help teachers explain how results were reached.

Rubric Criterion	Weight	What It Examines
Research & Evidence	30%	Relevant sources, supporting evidence and appropriate references.
Accuracy & Analysis	30%	Correctness, interpretation of results and depth of analysis.
Creativity & Application	25%	Originality, practical application and thoughtful problem-solving.
Communication	15%	Organization, clarity, explanation and presentation.

The concept describes a sample report of approximately 800–1,000 words with cited sources, experiment photographs or materials where relevant, a temperature data table or chart, and a reflection of approximately 100–250 words. Individual teachers may define different requirements for other projects.

Submission Rule	Specification in the Project Concept
Accepted formats	PDF, DOCX, PPTX, JPG and ZIP
Maximum per file	20 MB
Total submission size	50 MB
Required components	Report, supporting evidence, data table/chart and reflection when requested
Review sequence	Submission, preliminary AI review, teacher review, final publication

A rubric should be explained in student-friendly language. A high research score should be supported by evidence and citations, while a high communication score should reflect clarity and organization. A numerical score alone is less useful than a score accompanied by criterion-level comments.

8.7 AI-Assisted Evaluation and Human Oversight

The AI-assisted evaluation component provides an initial view of a submission. It may highlight missing components, potential gaps against rubric criteria and possible improvements. It should not replace a teacher’s understanding of learning objectives, classroom context or individual circumstances.

The system receives the submitted project and assignment requirements.

The AI assistance layer checks available evidence against the stated rubric and required components.

The system presents preliminary observations and areas that may need attention.

The teacher checks the original student work and determines whether the observations are appropriate.

The teacher assigns or confirms criterion-level scores, adds feedback and publishes the final result.

The student sees the published grade and feedback, with preliminary AI output clearly distinguished.

The interface reference gives an illustrative case in which the preliminary AI score is 86/100 and the teacher's score is 88/100 before publication. This example demonstrates why the two values should not be treated as interchangeable. The published teacher result is the final reviewed grade.

Students should understand whether a displayed score is preliminary or final.

Teachers should be able to disagree with an AI observation and record their own assessment.

AI suggestions should be related to assignment requirements and rubric criteria.

Career suggestions should be exploratory and not presented as guaranteed outcomes.

A single automated score should not be the sole basis for important academic decisions.

A future implementation should evaluate AI observations against teacher-reviewed examples, document limitations and provide a process for correcting inaccurate feedback.

8.8 Data Dictionary and Field Interpretation

The cleaned dataset contains 21 columns. The following field groups explain how the data can be interpreted for academic dashboards and career exploration.

Field	Interpretation	Possible Analytical Use
student_id	Student identifier.	Count records and connect project activity to a learner.
student_name	Student name field.	Display or group records where authorized; protect personal information.
age	Age value.	Describe the dataset population in aggregate.
city	City field.	Summarize records by location without assuming city determines ability.

education_level	Education category.	Compare records across education groups.
project_id	Project identifier.	Count unique projects and group records.
project_subject	Project subject/category.	Compare subject distributions and average scores.
skill_level	Beginner, Intermediate or Advanced.	Describe the level assigned to the project record.
programming_score	Programming competency score.	Compare programming with other competency measures.
analytical_score	Analytical competency score.	Review analytical performance patterns.
creativity_score	Creativity competency score.	Review creative application patterns.
communication_score	Communication competency score.	Review communication patterns.
problem_solving_score	Problem-solving competency score.	Review problem-solving patterns.
completeness_score	Completeness measure.	Identify whether submissions include expected components.
accuracy_score	Accuracy measure.	Review correctness-related patterns.
submitted_on_time	Submission timing indicator.	Calculate on-time and late-record proportions.
project_score	Project assessment score.	Summarize overall score distribution.
points_earned	Points associated with project activity.	Summarize gamification-related points.
ai_feedback	Qualitative feedback category.	Count common feedback themes.
career_interest	Recorded career interest.	Compare interest patterns.
recommended_career	Career recommendation field.	Summarize recommendations present in the dataset.

Fields should be interpreted according to their definitions and collection process. A recommended_career value records a recommendation in the dataset; it does not establish that a student has chosen that career or that the recommendation is objectively correct.

8.9 Expanded Dataset Findings

The dataset contains 1,000 student-project records, 21 columns, 100 unique project IDs, 10 project subjects and records from 10 cities. The findings below describe the supplied dataset and should not be generalized to all students without evidence that it is representative.

Measure	Observed Value	Interpretation
Average project score	69.79	Mean score across supplied records.
Median project score	69.8	Midpoint of the observed score distribution.
Minimum project score	35.8	Lowest score in the supplied dataset.
Maximum project score	96.5	Highest score in the supplied dataset.
Average points earned	94.33	Mean points value across records.
On-time records	808 / 1,000 (80.8%)	Most records are marked submitted on time.
Not-on-time records	192 / 1,000 (19.2%)	A smaller but meaningful group is marked not on time.
Intermediate skill level	458 records	Largest skill-level group.
Beginner skill level	401 records	Second-largest skill-level group.
Advanced skill level	141 records	Smallest skill-level group.

Average competency values are programming 57.61, analytical 61.31, creativity 61.68, communication 65.01, problem solving 67.08, completeness 77.30 and accuracy 61.29. Completeness is the highest average measure, while programming is the lowest. This difference can guide further investigation but does not by itself explain why scores differ.

A next analytical step would be to compare competency averages by subject, skill level and submission status, then inspect whether sample sizes are sufficient. Any association should be described as a pattern in the dataset rather than proof that one factor causes another.

8.10 Career Guidance Logic and Interpretation

Project Phoenix connects demonstrated project skills with recorded career interests and recommended careers. The purpose is to help students explore pathways and identify practical next steps. The dataset includes UI/UX Designer, Product Manager, Mobile App

Developer, Database Administrator, Data Analyst, Software Developer, Cloud Engineer, Cyber Security Analyst and AI/ML Engineer among its recommendations.

The most frequent recommended careers in the supplied records are UI/UX Designer (277), Product Manager (213), Mobile App Developer (117), Database Administrator (107) and Data Analyst (93). These counts describe the dataset; they do not indicate the number of available jobs in the labour market or prove that a career is best for every student.

Suggested Student-Facing Guidance Flow

Show the student a summary of reviewed projects and demonstrated competencies.

Display recorded career interests separately from system-generated recommendations.

Present exploratory pathways and explain which observed skills or interests relate to each one.

Recommend practical learning activities, such as a small project, a topic to study or a skill to practise.

Encourage discussion with a teacher or counsellor and revision of goals as new evidence appears.

A useful career guidance screen should avoid permanently classifying a student. Skills change with practice and interests can change through exposure to new subjects. Recommendations should therefore remain revisable and their supporting evidence should be understandable.

8.11 Dashboard and Reporting Design

Role-based dashboards reduce the need for users to interpret raw tables. The system presents information relevant to each user’s responsibilities.

Workspace	Important Dashboard Elements	Purpose
Student	Active projects, due dates, progress, reviewed grades, points, badges, feedback and career guidance.	Help students plan work and understand progress.
Teacher	Active assignments, class size, submission counts, pending reviews, evaluation status and class insights.	Help teachers prioritize reviews and identify patterns.
Parent	Progress summary, reviewed work, strengths, goals, pathways and counselling notes.	Support informed learning conversations.

Analytics	Subject distribution, competency averages, score distribution, submission timing and career counts.	Convert the dataset into interpretable summaries.

Dashboard design should prioritize important actions and avoid overwhelming users with every metric at once. For students, deadlines and actionable feedback may be more useful than a large chart. For teachers, pending reviews and class comparisons may be more useful than gamification metrics. For parents, progress summaries and agreed next steps may be more appropriate than technical measures.

Charts should have clear titles, readable labels and a note about the population summarized. When a chart uses the supplied 1,000-record dataset, it should be described as dataset analysis rather than as a representation of every student.

8.12 Testing Strategy and Acceptance Criteria

Testing should check whether the main workflow behaves as expected and whether each role receives the correct information. The scenarios below are proposed acceptance tests, not a claim that all tests have already been executed.

Test ID	Scenario	Expected Result
TC-01	Teacher creates and publishes an assignment.	Assignment appears in the student workspace with instructions and deadline.
TC-02	Student submits a permitted file.	Submission is associated with the correct assignment and status changes appropriately.
TC-03	Student attempts an unsupported file type.	The system communicates permitted formats and prevents or flags the invalid submission.
TC-04	AI preliminary evaluation is displayed.	It is labelled preliminary and not presented as a final grade.
TC-05	Teacher changes the preliminary assessment.	Teacher rubric scores and feedback determine the reviewed result.
TC-06	Teacher publishes the grade.	Published result becomes visible in the student progress view.

TC-07	Parent opens linked progress.	Only the linked student's permitted information is shown.
TC-08	Analytics summarizes the dataset.	Counts and averages correspond to the selected dataset and filters.
TC-09	Student opens career guidance.	Recommendations appear as exploratory guidance with context.
TC-10	Portfolio displays reviewed projects.	Portfolio distinguishes reviewed evidence from unreviewed drafts.

A production-ready system would also need security testing, access-control testing, file-upload validation, backup and recovery testing, usability evaluation and privacy review. Results should record test date, test data, actual outcome and any defect found.

8.13 Privacy, Security and Responsible Data Use

The application concept involves student records, submissions, grades, feedback and career-related information. These data should be handled carefully because they may influence students' understanding of progress and discussions about future goals.

Apply role-based access so students, teachers and parents access only records relevant to their roles.

Use secure authentication and protect sessions against unauthorized access.

Validate uploaded files, enforce size limits and restrict accepted file types.

Use secure transport for data exchanged between browser and application services.

Back up important records and establish a recovery process.

Keep preliminary AI feedback distinct from teacher-published grades.

Explain how data are used for dashboards and recommendations.

Limit unnecessary collection and retention of personal information.

Provide a way to correct inaccurate records and review disputed assessment information.

Avoid treating city, age or education category as a definitive measure of ability.

These points are recommended safeguards. The report does not claim that a formal security audit, privacy certification or production penetration test has been completed. Such activities would be required before deploying a platform with real student records at scale.

8.14 Limitations of the Current Prototype and Dataset

The report describes a project concept, interface references, a cleaned dataset and an academic prototype workflow. Several limitations should be acknowledged so that the results are not overstated.

The dataset contains 1,000 records, but its description does not establish that it is statistically representative of all students.

An association between a skill score and a recommended career does not prove that the recommendation is appropriate for an individual.

The supplied fields do not document the full method used to generate every career recommendation.

AI evaluation can miss context, misunderstand evidence or provide inaccurate suggestions; teacher review remains necessary.

Average values hide variation between individual students and projects.

A project score may not fully capture collaboration, persistence, creativity or growth over time.

Proposed security, audit and testing practices require implementation and verification before production use.

Dashboard accuracy depends on correct cleaning, field definitions and filtering.

These limitations suggest iterative development. Future versions should document data provenance, validate recommendation logic, evaluate AI feedback against teacher judgement and gather feedback from students, teachers and parents.

8.15 Future Enhancements and Development Plan

Future development can improve the project in stages. The priorities follow the central purpose of Project Phoenix: make project work easier to manage, make feedback more useful and connect reviewed learning evidence to career exploration.

Stage	Enhancement	Expected Benefit
Stage 1	Stabilize assignment creation, submission status, teacher review and grade publication.	Reliable end-to-end academic workflow.
Stage 2	Improve dashboards, filters, dataset summaries and downloadable reports.	Clearer progress visibility.
Stage 3	Add portfolio evidence linked to reviewed projects and competencies.	Stronger record of student work and skills.

Stage 4	Make career recommendations explainable and revisable.	More transparent pathway exploration.
Stage 5	Add audit trails, access reviews, backups and documented privacy controls.	Better accountability and protection.
Stage 6	Evaluate the interface with students, teachers and parents.	Usability improvements based on feedback.
Stage 7	Validate AI observations against teacher-reviewed examples and document errors.	More responsible AI assistance.

The stages are a proposed development plan rather than a confirmed delivery schedule. Priorities should be adjusted according to implementation status, available resources and feedback from the project guide.

8.16 Summary of Extended Technical Discussion

The extended analysis shows how Project Phoenix modules can form a coherent academic workflow. Functional requirements define actions expected from students, teachers and parents. Non-functional requirements describe usability, reliability, privacy and transparency goals. Use cases and acceptance tests provide concrete ways to review the workflow.

The dataset adds a quantitative view of student-project records, including project scores, submission status, competency measures, feedback categories and career recommendations. These figures can support exploratory dashboards, but their interpretation must remain limited to the supplied data. Career recommendations should be explained and revisited as students gain experience.

Most importantly, the proposed system maintains a clear distinction between preliminary AI assistance and final teacher judgement. By combining project management, reviewed feedback, portfolio evidence, analytics and career exploration, Project Phoenix offers a structured foundation for supporting student development while leaving meaningful educational decisions with teachers and students.

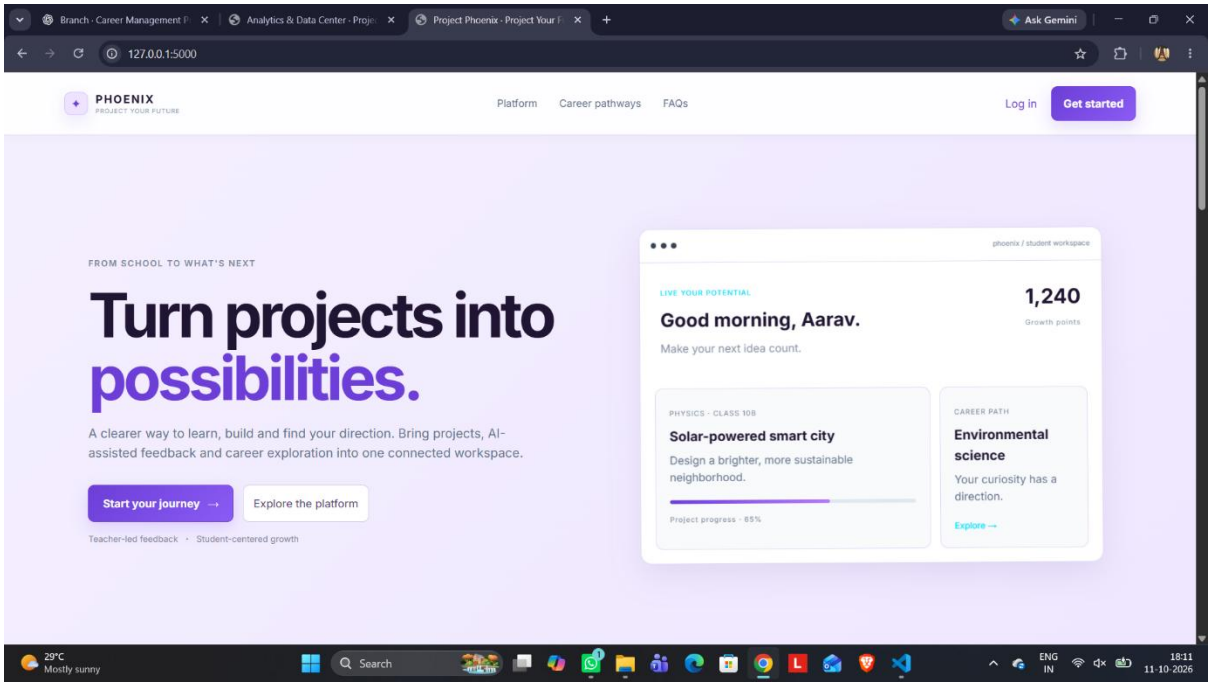
9. Output

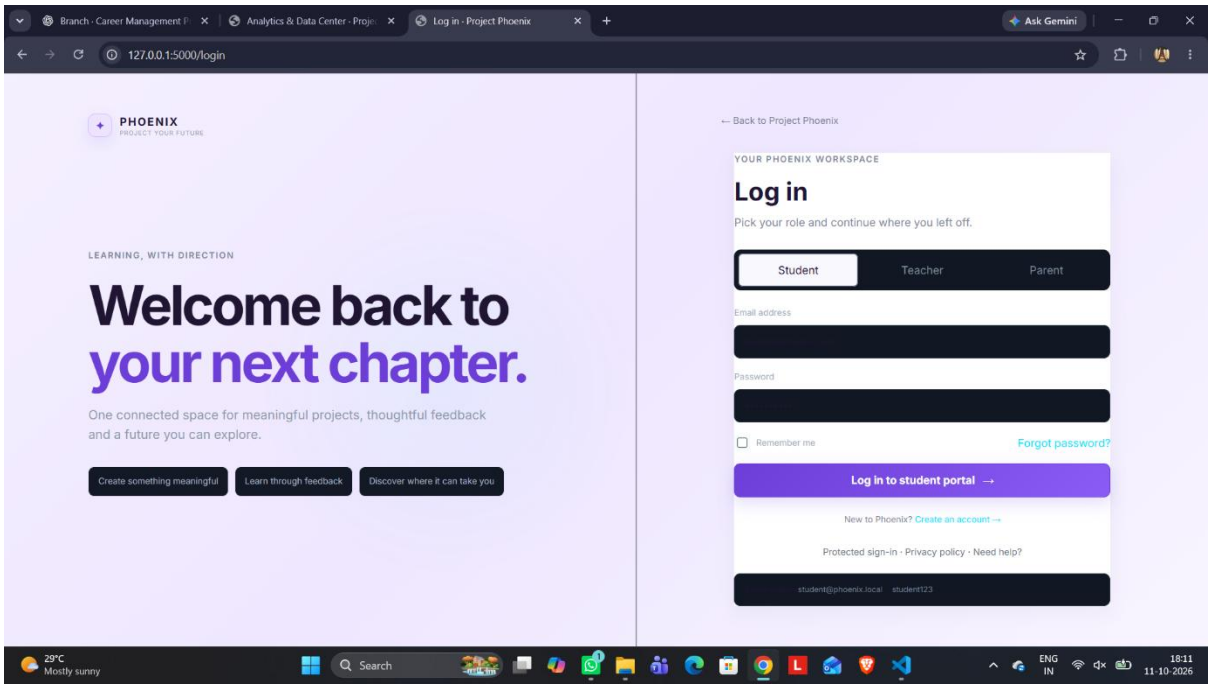
The Project Phoenix prototype provides different outputs for each role. The UI reference document presents a shared portal shell and role-specific workspaces, while the project presentation defines the functional workflow around project management, evaluation, career exploration, portfolio, and PTM counselling.

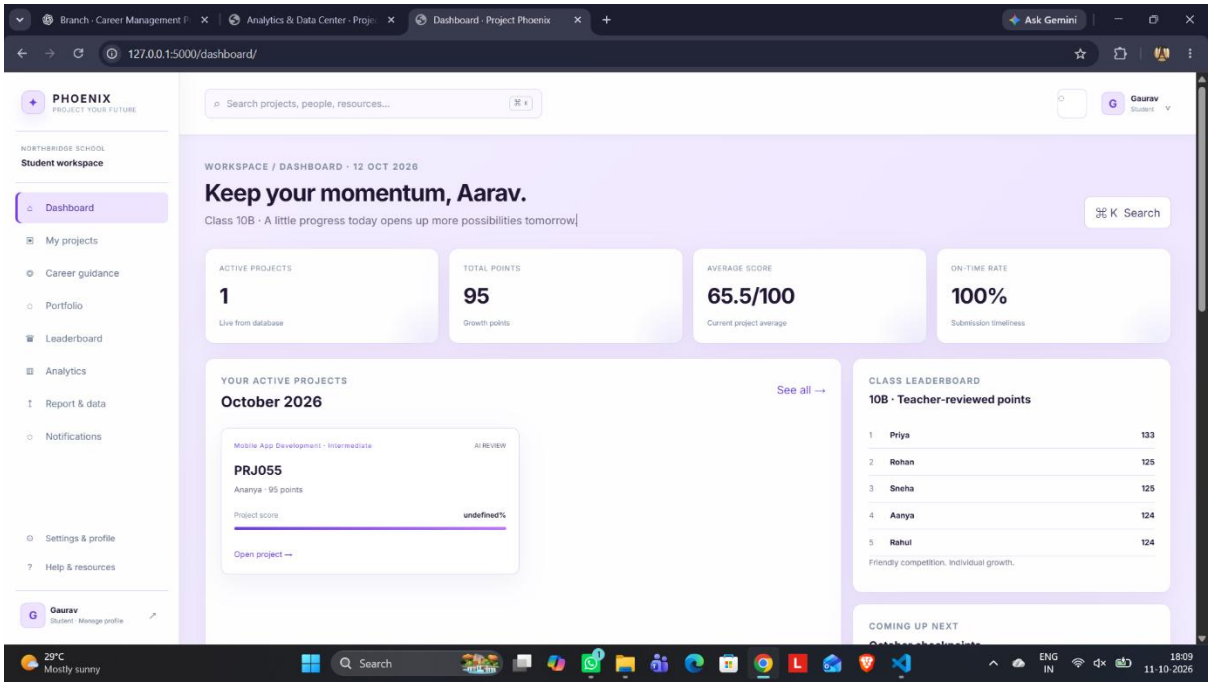
Output / Screen	Main Information Displayed
Student Dashboard	Active projects, deadlines, progress, total points, class leaderboard, badges, feedback, and upcoming checkpoints.
My Projects / Submission	Project requirements, files/links, submission status, reflection, and project progress.
Teacher Dashboard	Class overview, active assignments, submission counts, pending reviews, and class-level indicators.
Assignment Creation	Project brief, class/subject, skill level, deadline, requirements, accepted file types, rubric, and publishing checklist.
Evaluation Inbox	Student submission, preliminary AI score, rubric criteria, teacher scores, final feedback, and review status.
Analytics / Class Insights	Project-score distributions, competency comparisons, submission status, subject analysis, and career-related patterns.
Career Guidance	Skill profile, career interests, suggested pathways, explanation of the match, and next skills to develop.
Portfolio / Leaderboard	Teacher-reviewed projects, demonstrated skills, points, badges, class rank, and selected portfolio evidence.
PTM Counselling	Progress summary, strengths, goals, career pathways, and discussion points for parent-teacher meetings.

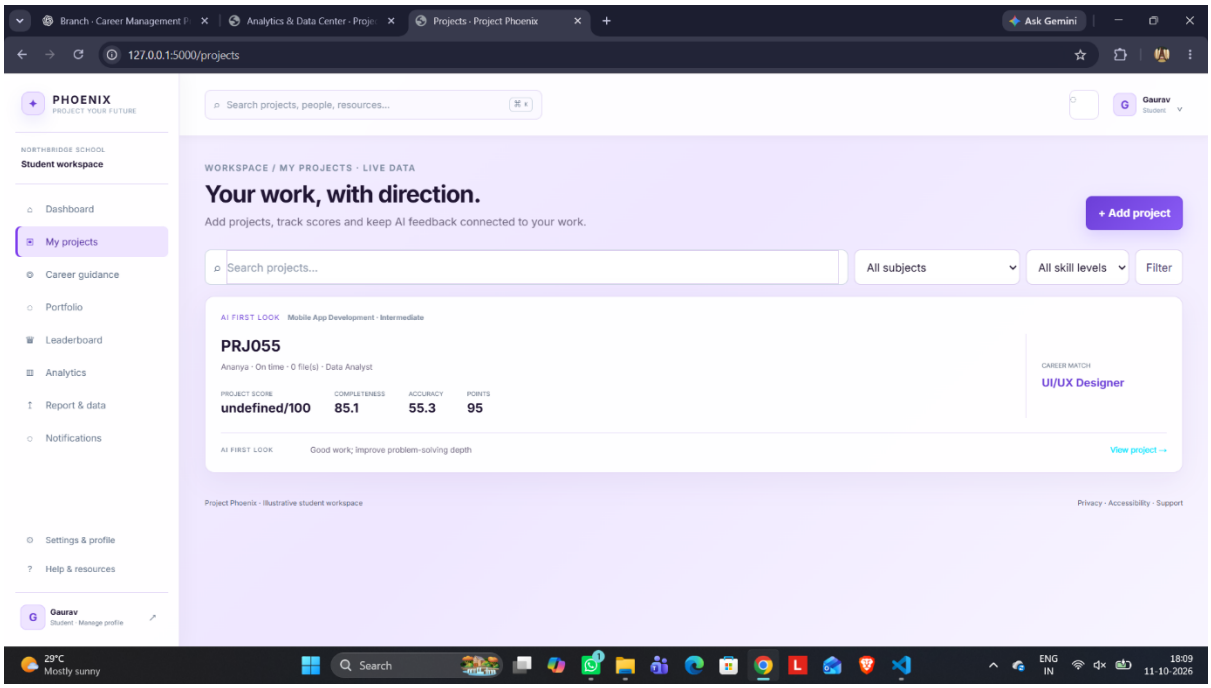
The UI reference includes examples of a student dashboard with active projects, points, class ranking, badges and feedback, and a teacher evaluation inbox where AI scores are explicitly preliminary and the teacher publishes the final grade.

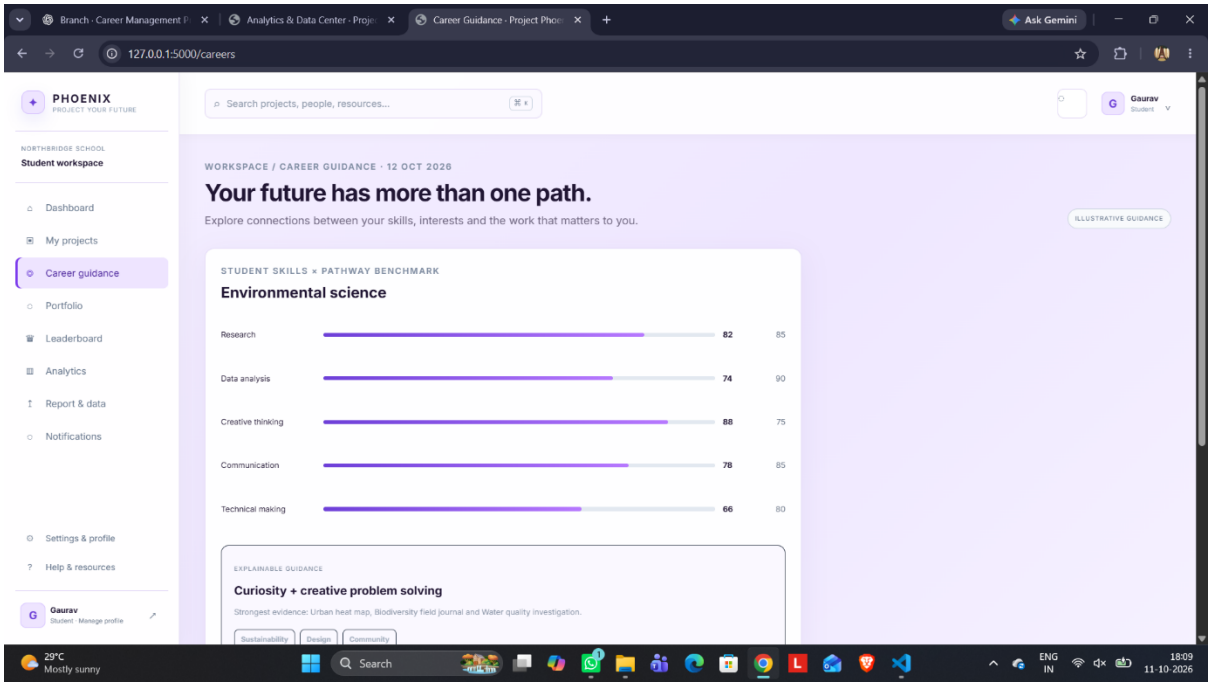
The proposed output therefore connects raw student/project data to understandable academic views. It also separates AI assistance from teacher authority, which is important for responsible academic evaluation.

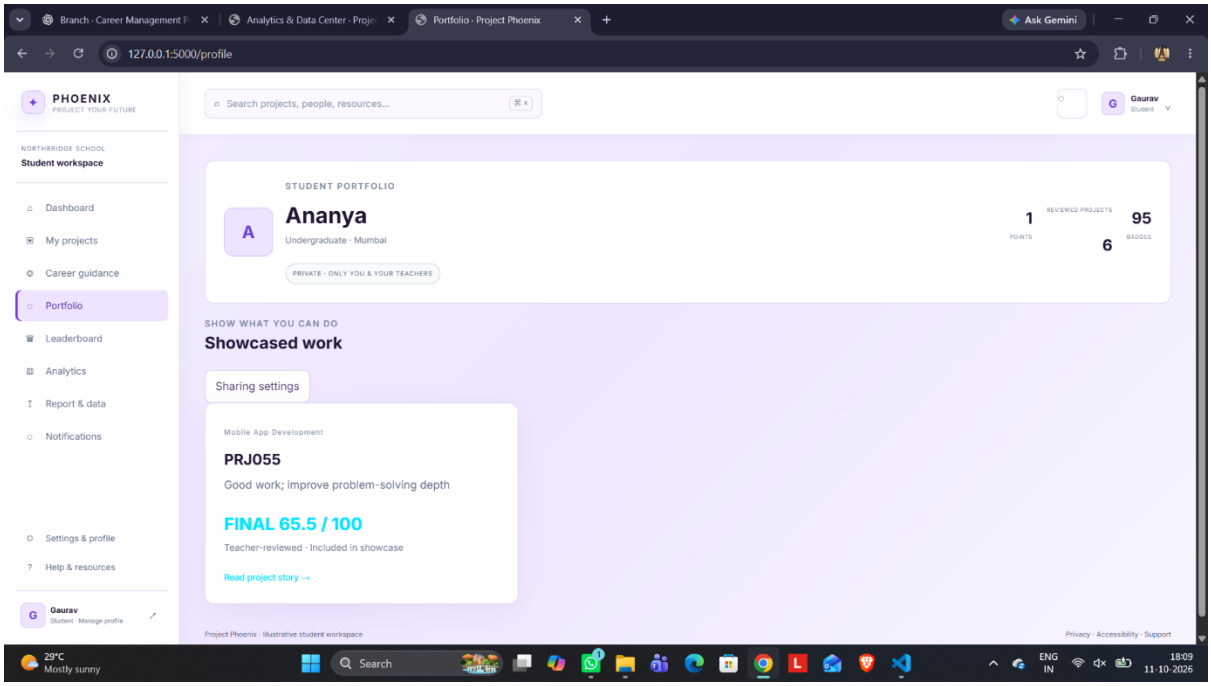

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Project Phoenix · Project Your F
Ask Gemini
127.0.0.1:5000
PHOENIX
PROJECT YOUR FUTURE
Platform
Career pathways
FAQs
Log in
Get started
FROM SCHOOL TO WHAT'S NEXT
Turn projects into possibilities.
A clearer way to learn, build and find your direction. Bring projects, AI-assisted feedback and career exploration into one connected workspace.
Start your journey →
Explore the platform
Teacher-led feedback · Student-centered growth
phoenix / student workspace
LIVE YOUR POTENTIAL
Good morning, Aarav.
Make your next idea count.
1,240
Growth points
PHYSICS · CLASS 10B
Solar-powered smart city
Design a brighter, more sustainable neighborhood.
Project progress · 65%
CAREER PATH
Environmental science
Your curiosity has a direction.
Explore →
29°C
Mostly sunny
Search
ENG
IN
18:11
11-10-2026

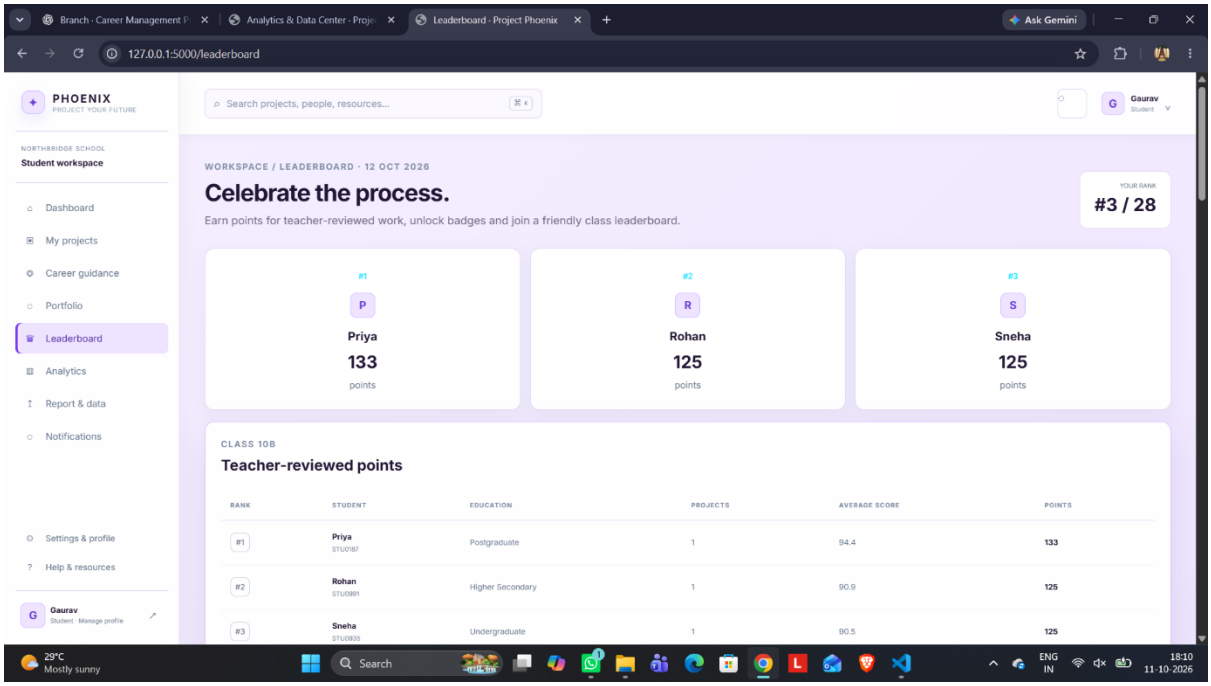

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Log in · Project Phoenix
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127.0.0.1:5000/login
PHOENIX
PROJECT YOUR FUTURE
LEARNING, WITH DIRECTION
Welcome back to your next chapter.
One connected space for meaningful projects, thoughtful feedback and a future you can explore.
Create something meaningful
Learn through feedback
Discover where it can take you
← Back to Project Phoenix
YOUR PHOENIX WORKSPACE
Log in
Pick your role and continue where you left off.
Student
Teacher
Parent
Email address
Password
Remember me
Forgot password?
Log in to student portal →
New to Phoenix? Create an account →
Protected sign-in · Privacy policy · Need help?
student@phoenix.local
student123
29°C
Mostly sunny
Search
ENG
IN
18:11
11-10-2026

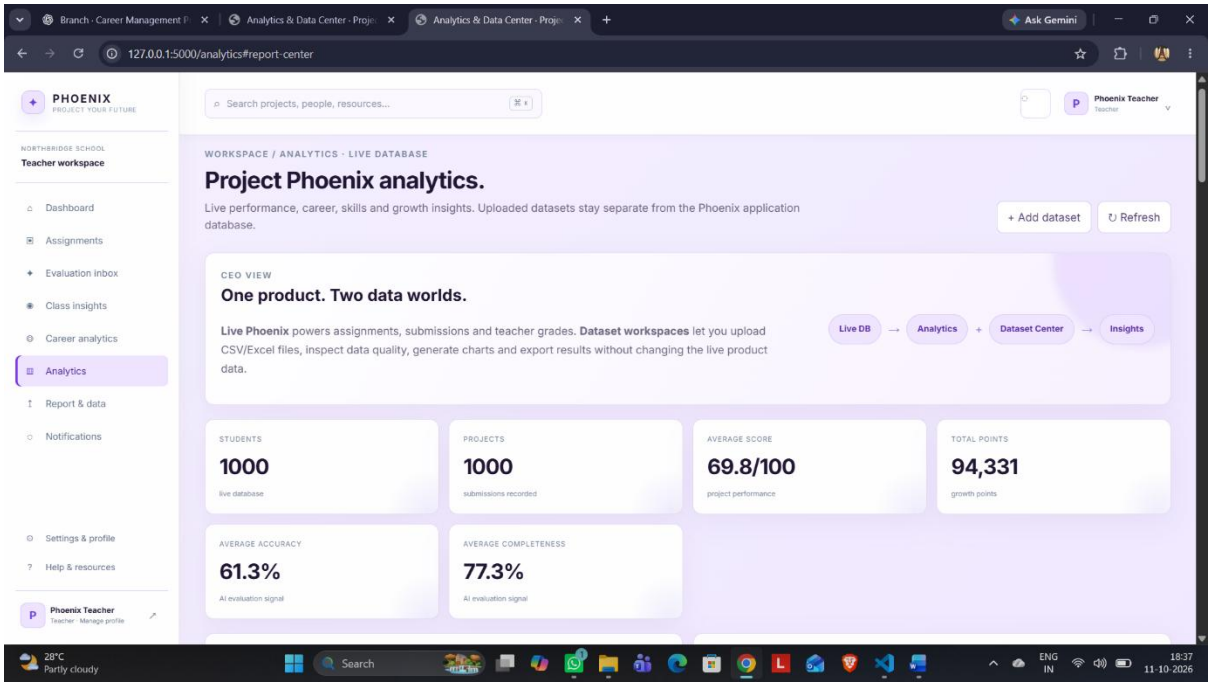

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Dashboard · Project Phoenix
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PHOENIX
PROJECT YOUR FUTURE
Search projects, people, resources...
Gaurav
Student
NORTHBRIDGE SCHOOL
Student workspace
Dashboard
My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
WORKSPACE / DASHBOARD · 12 OCT 2026
Keep your momentum, Aarav.
Class 10B · A little progress today opens up more possibilities tomorrow.
⌘K Search
ACTIVE PROJECTS
1
Live from database
TOTAL POINTS
95
Growth points
AVERAGE SCORE
65.5/100
Current project average
ON-TIME RATE
100%
Submission timeliness
YOUR ACTIVE PROJECTS
October 2026
See all →
Mobile App Development · Intermediate
AI REVIEW
PRJ055
Ananya · 95 points
Project score
undefined%
Open project →
CLASS LEADERBOARD
10B · Teacher-reviewed points
1 Priya 133
2 Rohan 125
3 Sneha 125
4 Aanya 124
5 Rahul 124
Friendly competition. Individual growth.
COMING UP NEXT
29°C
Mostly sunny
Search
ENG
IN
18:09
11-10-2026

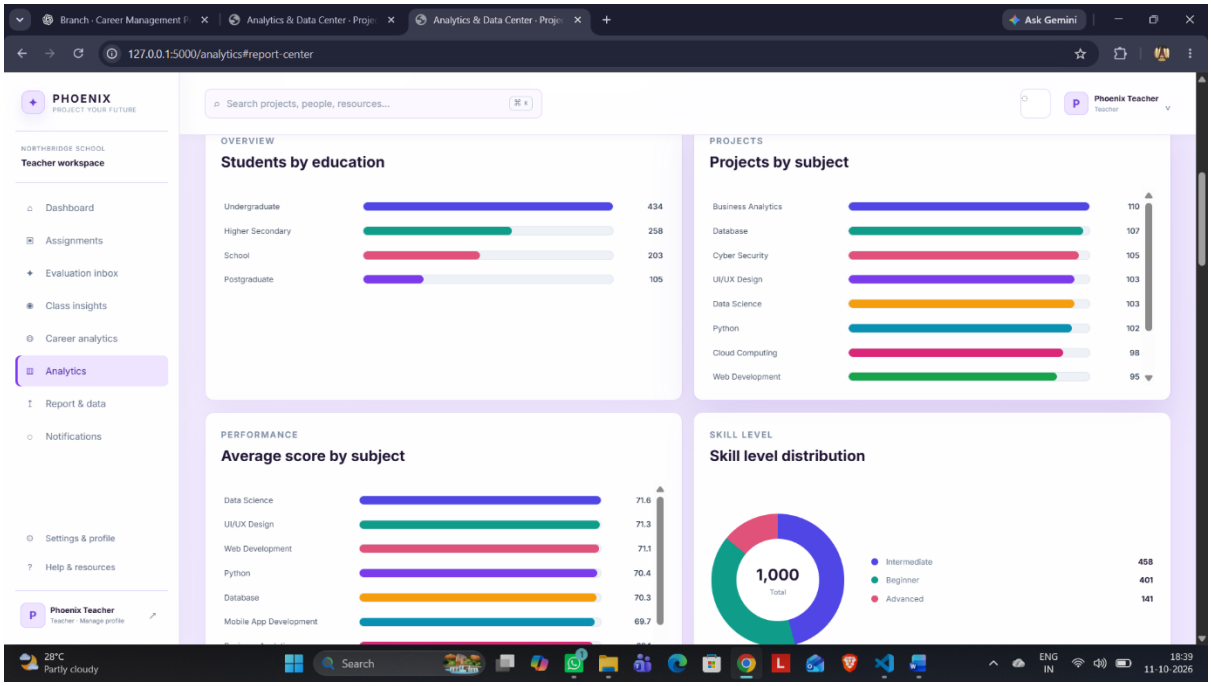

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Projects · Project Phoenix
Ask Gemini
127.0.0.1:5000/projects
PHOENIX
PROJECT YOUR FUTURE
Search projects, people, resources...
Gaurav
Student
NORTHBRIDGE SCHOOL
Student workspace
Dashboard
My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
WORKSPACE / MY PROJECTS · LIVE DATA
Your work, with direction.
Add projects, track scores and keep AI feedback connected to your work.
+ Add project
Search projects...
All subjects
All skill levels
Filter
AI FIRST LOOK
Mobile App Development · Intermediate
PRJ055
Ananya · On time · 0 file(s) · Data Analyst
PROJECT SCORE
undefined/100
COMPLETENESS
85.1
ACCURACY
55.3
POINTS
95
CAREER MATCH
UI/UX Designer
AI FIRST LOOK
Good work; improve problem-solving depth
View project →
Project Phoenix · Illustrative student workspace
Privacy · Accessibility · Support
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Search
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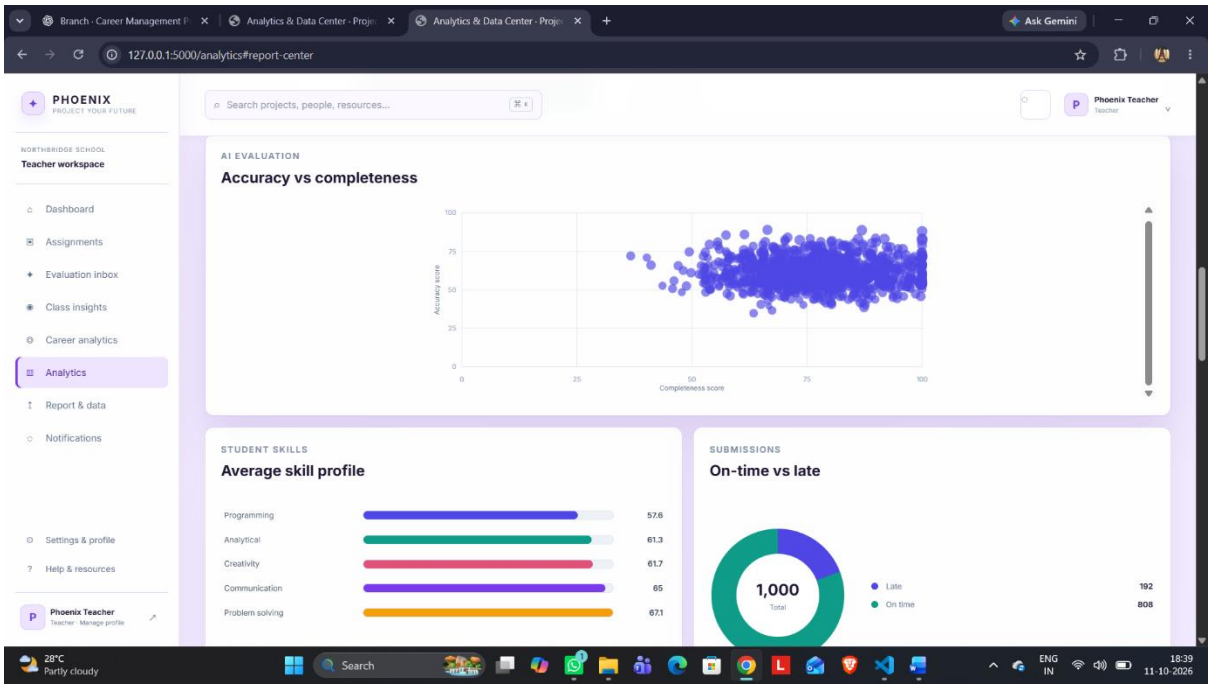

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Search projects, people, resources...
Gaurav
Student
NORTHBRIDGE SCHOOL
Student workspace
Dashboard
My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
WORKSPACE / CAREER GUIDANCE · 12 OCT 2026
Your future has more than one path.
Explore connections between your skills, interests and the work that matters to you.
ILLUSTRATIVE GUIDANCE
STUDENT SKILLS × PATHWAY BENCHMARK
Environmental science
Research 82 85
Data analysis 74 90
Creative thinking 88 75
Communication 78 85
Technical making 66 80
EXPLAINABLE GUIDANCE
Curiosity + creative problem solving
Strongest evidence: Urban heat map, Biodiversity field journal and Water quality investigation.
Sustainability
Design
Community
29°C
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Search
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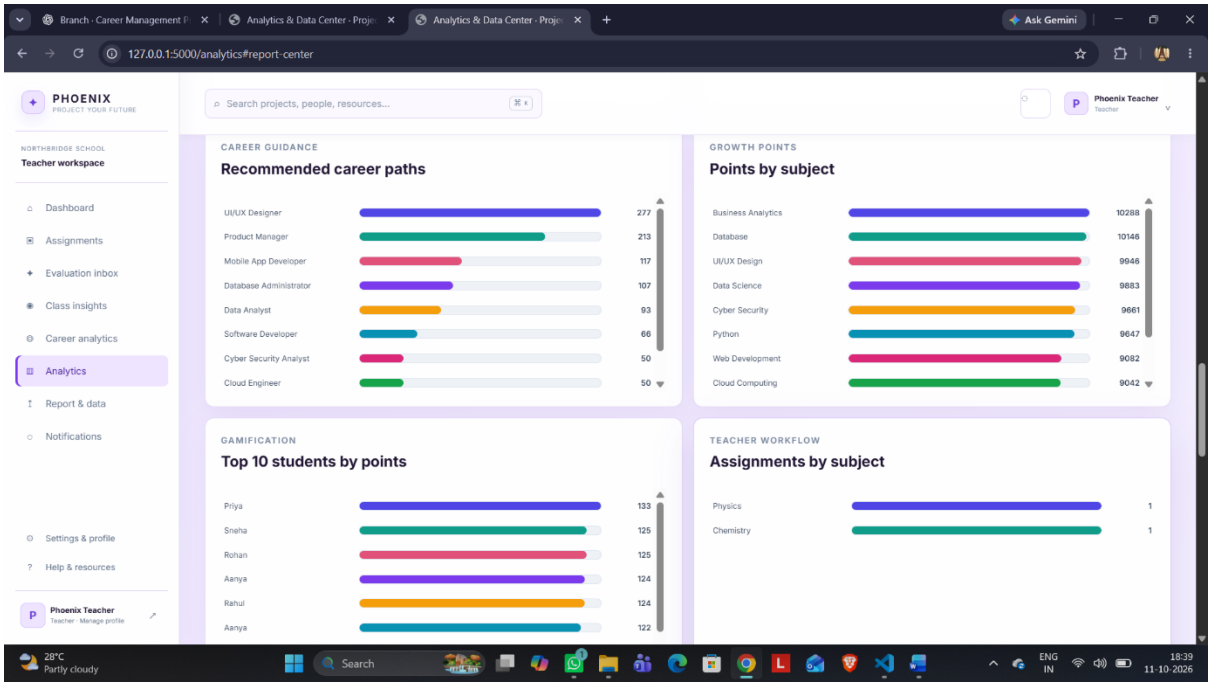

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Portfolio · Project Phoenix
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PHOENIX
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Search projects, people, resources...
Gaurav
Student
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Student workspace
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My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
STUDENT PORTFOLIO
A
Ananya
Undergraduate · Mumbai
PRIVATE - ONLY YOU & YOUR TEACHERS
1
REVIEWED PROJECTS
95
POINTS
6
BADGES
SHOW WHAT YOU CAN DO
Showcased work
Sharing settings
Mobile App Development
PRJ055
Good work; improve problem-solving depth
FINAL 65.5 / 100
Teacher-reviewed · Included in showcase
Read project story →
Project Phoenix · Illustrative student workspace
Privacy · Accessibility · Support
29°C
Mostly sunny
Search
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11-10-2026

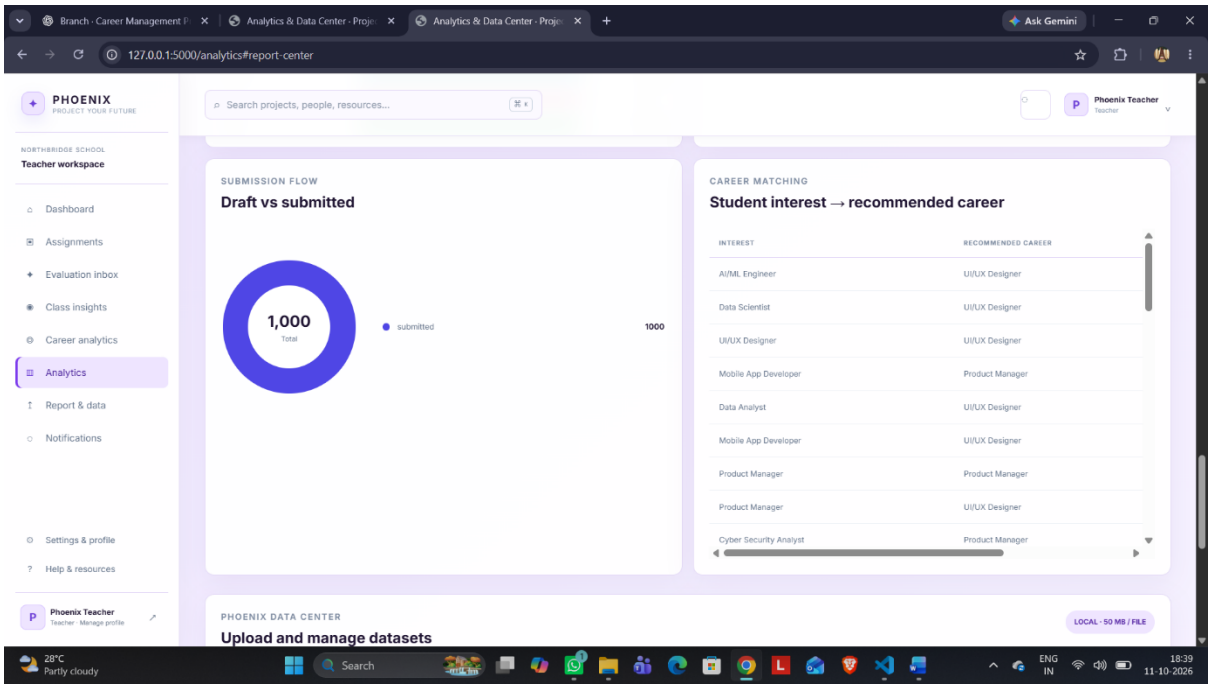

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Analytics & Data Center - Proje
Leaderboard - Project Phoenix
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PHOENIX
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Search projects, people, resources...
Gaurav
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NORTHBRIDGE SCHOOL
Student workspace
Dashboard
My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
WORKSPACE / LEADERBOARD · 12 OCT 2026
Celebrate the process.
Earn points for teacher-reviewed work, unlock badges and join a friendly class leaderboard.
YOUR RANK
#3 / 28
#1
P
Priya
133
points
#2
R
Rohan
125
points
#3
S
Sneha
125
points
CLASS 10B
Teacher-reviewed points
RANK
STUDENT
EDUCATION
PROJECTS
AVERAGE SCORE
POINTS
#1
Priya
STU0187
Postgraduate
1
94.4
133
#2
Rohan
STU0991
Higher Secondary
1
90.9
125
#3
Sneha
STU0935
Undergraduate
1
90.5
125
29°C
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Search
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18:10
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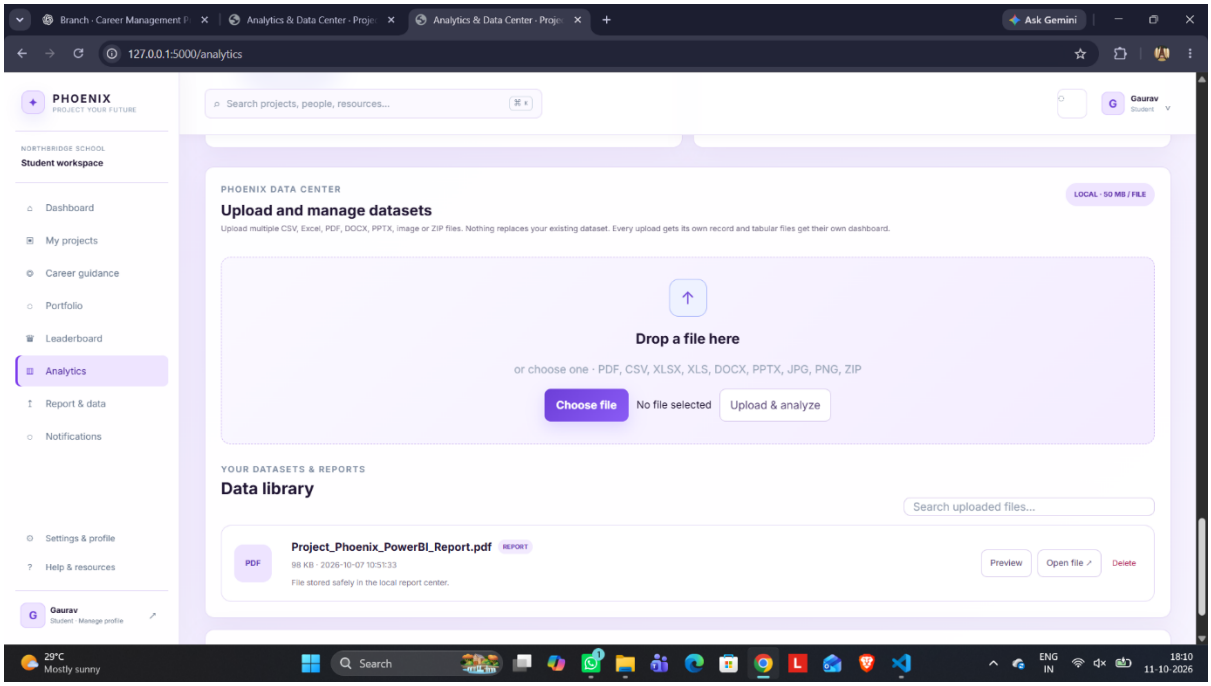

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PHOENIX
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Search projects, people, resources...
Phoenix Teacher
Teacher
NORTHBRIDGE SCHOOL
Teacher workspace
Dashboard
Assignments
Evaluation inbox
Class insights
Career analytics
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Phoenix Teacher
Teacher · Manage profile
WORKSPACE / ANALYTICS · LIVE DATABASE
Project Phoenix analytics.
Live performance, career, skills and growth insights. Uploaded datasets stay separate from the Phoenix application database.
+ Add dataset
↻ Refresh
CEO VIEW
One product. Two data worlds.
Live Phoenix powers assignments, submissions and teacher grades. Dataset workspaces let you upload CSV/Excel files, inspect data quality, generate charts and export results without changing the live product data.
Live DB
Analytics
Dataset Center
Insights
STUDENTS
1000
live database
PROJECTS
1000
submissions recorded
AVERAGE SCORE
69.8/100
project performance
TOTAL POINTS
94,331
growth points
AVERAGE ACCURACY
61.3%
AI evaluation signal
AVERAGE COMPLETENESS
77.3%
AI evaluation signal
28°C
Partly cloudy
Search
ENG
IN
18:37
11-10-2026

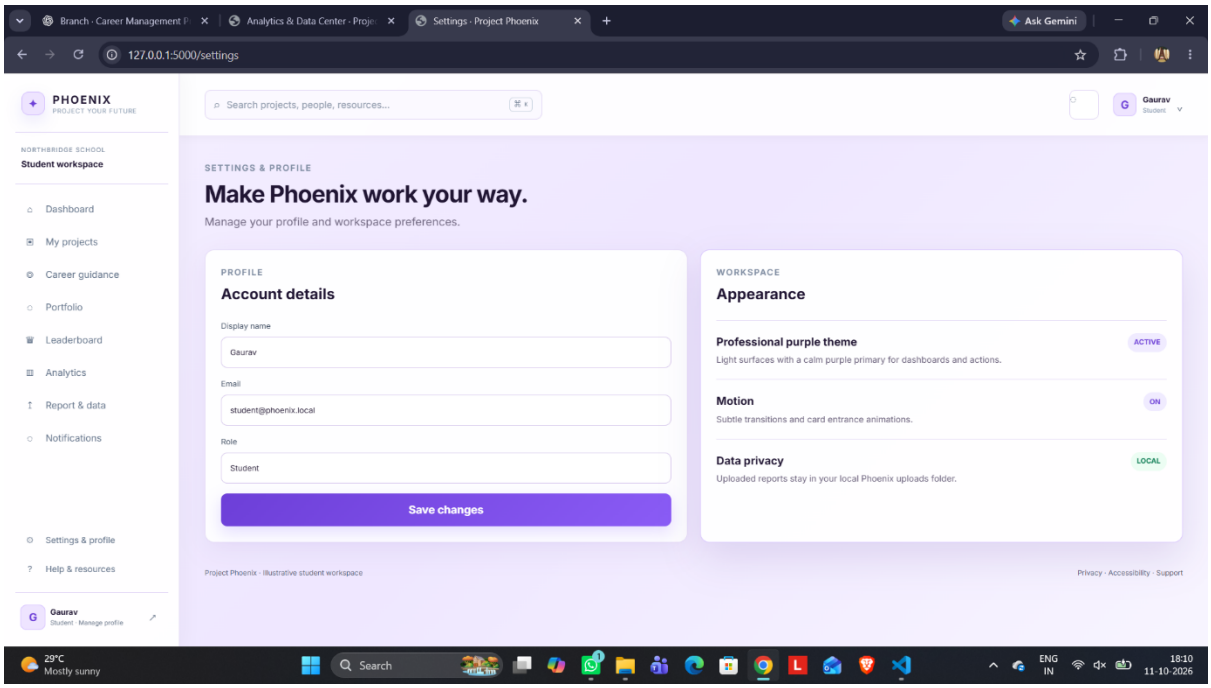

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Analytics & Data Center - Proje
Analytics & Data Center - Proje
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PHOENIX
PROJECT YOUR FUTURE
Search projects, people, resources...
Phoenix Teacher
Teacher
NORTHBRIDGE SCHOOL
Teacher workspace
Dashboard
Assignments
Evaluation inbox
Class insights
Career analytics
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Phoenix Teacher
Teacher · Manage profile
OVERVIEW
Students by education
Undergraduate 434
Higher Secondary 258
School 203
Postgraduate 105
PROJECTS
Projects by subject
Business Analytics 110
Database 107
Cyber Security 105
UI/UX Design 103
Data Science 103
Python 102
Cloud Computing 98
Web Development 95
PERFORMANCE
Average score by subject
Data Science 71.6
UI/UX Design 71.3
Web Development 71.1
Python 70.4
Database 70.3
Mobile App Development 69.7
SKILL LEVEL
Skill level distribution
1,000
Total
Intermediate 458
Beginner 401
Advanced 141
28°C
Partly cloudy
Search
ENG
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11-10-2026


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Analytics & Data Center - Proje
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PHOENIX
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Search projects, people, resources...
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Teacher
NORTHBRIDGE SCHOOL
Teacher workspace
Dashboard
Assignments
Evaluation inbox
Class insights
Career analytics
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Phoenix Teacher
Teacher · Manage profile
AI EVALUATION
Accuracy vs completeness
Accuracy score
Completeness score
STUDENT SKILLS
Average skill profile
Programming 57.6
Analytical 61.3
Creativity 61.7
Communication 65
Problem solving 67.1
SUBMISSIONS
On-time vs late
1,000
Total
Late 192
On time 808
28°C
Partly cloudy
Search
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18:39
11-10-2026


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Analytics & Data Center · Proje
Analytics & Data Center · Proje
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127.0.0.1:5000/analytics#report-center
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Phoenix Teacher
Teacher
NORTHBRIDGE SCHOOL
Teacher workspace
Dashboard
Assignments
Evaluation inbox
Class insights
Career analytics
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Phoenix Teacher
Teacher · Manage profile
CAREER GUIDANCE
Recommended career paths
UI/UX Designer 277
Product Manager 213
Mobile App Developer 117
Database Administrator 107
Data Analyst 93
Software Developer 66
Cyber Security Analyst 50
Cloud Engineer 50
GROWTH POINTS
Points by subject
Business Analytics 10288
Database 10146
UI/UX Design 9946
Data Science 9883
Cyber Security 9861
Python 9647
Web Development 9082
Cloud Computing 9042
GAMIFICATION
Top 10 students by points
Priya 133
Sneha 125
Rohan 125
Aanya 124
Rahul 124
Aanya 122
TEACHER WORKFLOW
Assignments by subject
Physics 1
Chemistry 1
28°C
Partly cloudy
Search
ENG
IN
18:39
11-10-2026


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Analytics & Data Center · Proje
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Search projects, people, resources...
Phoenix Teacher
Teacher
NORTHBRIDGE SCHOOL
Teacher workspace
Dashboard
Assignments
Evaluation inbox
Class insights
Career analytics
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Phoenix Teacher
Teacher · Manage profile
SUBMISSION FLOW
Draft vs submitted
1,000
Total
submitted 1000
CAREER MATCHING
Student interest → recommended career
INTEREST
RECOMMENDED CAREER
AI/ML Engineer UI/UX Designer
Data Scientist UI/UX Designer
UI/UX Designer UI/UX Designer
Mobile App Developer Product Manager
Data Analyst UI/UX Designer
Mobile App Developer UI/UX Designer
Product Manager Product Manager
Product Manager UI/UX Designer
Cyber Security Analyst Product Manager
PHOENIX DATA CENTER
Upload and manage datasets
LOCAL · 50 MB / FILE
28°C
Partly cloudy
Search
ENG
IN
18:39
11-10-2026


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Analytics & Data Center - Proje
Analytics & Data Center - Proje
Ask Gemini
127.0.0.1:5000/analytics
PHOENIX
PROJECT YOUR FUTURE
Search projects, people, resources...
Gaurav
Student
NORTHBRIDGE SCHOOL
Student workspace
Dashboard
My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
PHOENIX DATA CENTER
LOCAL · 50 MB / FILE
Upload and manage datasets
Upload multiple CSV, Excel, PDF, DOCX, PPTX, image or ZIP files. Nothing replaces your existing dataset. Every upload gets its own record and tabular files get their own dashboard.
Drop a file here
or choose one · PDF, CSV, XLSX, XLS, DOCX, PPTX, JPG, PNG, ZIP
Choose file
No file selected
Upload & analyze
YOUR DATASETS & REPORTS
Data library
Search uploaded files...
PDF
Project_Phoenix_PowerBI_Report.pdf
REPORT
98 KB · 2026-10-07 10:51:33
File stored safely in the local report center.
Preview
Open file
Delete
29°C
Mostly sunny
Search
ENG
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18:10
11-10-2026


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Settings · Project Phoenix
Ask Gemini
127.0.0.1:5000/settings
PHOENIX
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Search projects, people, resources...
Gaurav
Student
NORTHBRIDGE SCHOOL
Student workspace
Dashboard
My projects
Career guidance
Portfolio
Leaderboard
Analytics
Report & data
Notifications
Settings & profile
Help & resources
Gaurav
Student · Manage profile
SETTINGS & PROFILE
Make Phoenix work your way.
Manage your profile and workspace preferences.
PROFILE
Account details
Display name
Gaurav
Email
student@phoenix.local
Role
Student
Save changes
WORKSPACE
Appearance
Professional purple theme
ACTIVE
Light surfaces with a calm purple primary for dashboards and actions.
Motion
ON
Subtle transitions and card entrance animations.
Data privacy
LOCAL
Uploaded reports stay in your local Phoenix uploads folder.
Project Phoenix · Illustrative student workspace
Privacy · Accessibility · Support
29°C
Mostly sunny
Search
ENG
IN
18:10
11-10-2026

10. Conclusion

Project Phoenix demonstrates how a student project-management platform can be extended into a broader student development system. Instead of limiting the application to assignment submission and marks, the proposed system connects project work with feedback, skill evidence, achievements, analytics, career exploration, portfolio development, and parent-teacher counselling.

The dataset strengthens this approach by providing structured evidence across 1,000 student-project records. It contains project subjects, skill levels, seven competency measures, project scores, points, submission status, AI feedback, career interests, and recommended careers. This makes it suitable for dashboards, class insights, performance analysis, and exploratory career guidance.

The project presentation emphasizes centralized project management, AI-assisted evaluation, career exploration, gamified learning, portfolio development, and PTM counselling. The final system should continue to treat AI as an assistive layer rather than the sole decision-maker: teacher review remains central to final grading, while career recommendations should be interpreted as guidance rather than guaranteed predictions.

Overall, Project Phoenix provides a practical foundation for a unified academic platform in which students can build skills through meaningful projects, teachers can evaluate and guide more effectively, and parents can participate in informed progress and career discussions.

Source Materials Used for This Report

- Alliance University report-format document supplied for this project.
- Project Phoenix project-management presentation supplied for the project.
- Project Phoenix cleaned dataset (project_phoenix_cleaned.csv).
- Project Phoenix UI/reference document supplied for the website design and workflow.